

Mark schemes

1. A

[1]

2. D

[1]

3. Relevant issues raised:

Explanation

For a relevant explanation of supernormal profit eg in terms of profit in excess of normal profit which is the minimum return required to keep an enterprise in its current use.

Analysis of effects

For Example

- If firms are colluding closely, then the industry can be analysed as a (complex) monopoly. If the business is profit maximising then $MC = MR$; a percentage tax rate takes a proportion of total profit which is the difference between TC and TR. The rational business decision is to continue at the same level of output and price because a proportion of a maximised profit is better than a proportion of a less than maximum profit. The tax therefore has no immediate or short run effect on the equilibrium position of the firm, instead it acts as a revenue-raiser for the government; however, in the longer term the firm might react with decisions about such matters as investment or expansion.
- Other considerations, e.g. the special circumstance of companies that, in effect, sell electricity to themselves (a form of 'transfer pricing') and/or have to estimate demand in advance, and have a product that cannot be stored.
- Analysis in terms of supply and demand curves might be used to illustrate the effects of a tax. While this might not be appropriate to an imperfect market, which is the context of the data, the question does not specify the type of market. Also, it could (conceivably) be argued that in a global context, energy supply might be structured monopolistically but behave competitively, e.g. by price taking. While this might be more credible in the case of supermarkets, we will accept an argument that the tax will shift supply to the left and further raise prices, and mark the subsequent logical chain of reasoning on its quality.

For making appropriate use of diagrams, e.g. profits diagram for a monopolist; S & D analysis (see notes above).

[9]

4. B

[1]

7.

C

[1]

8.

Areas for discussion include:

- the meaning of cost minimisation/revenue maximisation
- ultimately, minimising total costs means not producing anything
- cost minimisation means minimising the cost of producing any given level of output
- X- inefficiency
- methods of reducing/minimising costs
- implications of cost minimisation
- methods of increasing/maximising revenues
- implications of revenue maximisation
- discussion of whether both can be pursued at once
- profit maximisation/satisficing
- explanation that while $\text{Total Profit} = \text{TR} - \text{TC}$, maximum profit is at $\text{MC} = \text{MR}$
- is cost minimisation and revenue maximisation consistent with profit maximisation?
- the need for firms in the private sector to make profits
- other possible objectives that firms might pursue
- the importance of the type of firm and the structure of the industry
- issues to do with the divorce of ownership and control; different sections/interests within a single firm might have different objectives, eg production process (minimise costs) sales/marketing (maximise total revenue) managers (utility/status/bonuses linked to profits/sales/other indices), shareholders (long-run, short-run perspectives on share prices and dividends)
- picking up on the word 'always' in the question

The use of relevant diagrams to support the analysis should be taken into account when assessing the quality of the candidate's response to the question.

[25]

9.

Areas for discussion include:

- what is meant by contestable markets
- nature of entry/exit barriers
- why lack of contestability leads to market power
- outcomes in terms of conduct/performance if markets are more contestable
- whether “perfect” contestability is possible
- hit and run competition
- examples of contestable markets
- role of technology is making markets more contestable
- how technology may make markets less contestable
- natural monopoly arguments
- role of regulators/state in creating contestability
- effectiveness of policies to create greater contestability
- whether contestability will solve the problems of potential market power
- the need for other forms of intervention to correct the possible abuses of market power
- other forms of government intervention and their effectiveness

The use of relevant diagrams to support the analysis should be taken into account when assessing the quality of the candidate's response to the question.

[25]

10.

In this part of the question, candidates will need to demonstrate that they are able to evaluate issues and arguments to support a conclusion if they are to be awarded **more than 15 marks**.

Where there is no **explicit** reference to the data, award a maximum of **21 marks**.

Level 5	Good analysis <u>and</u> good evaluation	22 to 25 marks <i>Mid-Point 24 marks</i>
Level 4	Good analysis <u>but</u> limited evaluation OR Reasonable analysis <u>and</u> reasonable evaluation	17 to 21 <i>Mid-Point 19 marks</i>
Level 3	Reasonable including some correct analysis <u>but</u> very limited evaluation	10 to 16 <i>Mid-Point 13 marks</i>
Level 2	Weak with some understanding	4 to 9 <i>Mid-Point 7 marks</i>
Level 1	Very weak	0 to 3 <i>Mid-Point 2 marks</i>

Refer also to the issues and areas for discussion on the next page.

Introduction	- Definitions / descriptions of, e.g. - Market forces - Government intervention / regulation
Developing the response to the question: (Application)	Comments from own economic knowledge or based on the data, e.g. - Further comments on Extract A, (international comparisons of green / non-green energy sources) - Examples from Extract B of businesses behaving responsibly, e.g seeking new fuels for aircraft, developing electric cars - Market forces clashing with government predictions, e.g. seeking new oil while wanting to limit climate change (Extract B) - Failures of regulation (Extract C) - Unsustainable business model (Extract C)
Developing the response to the question: (Analysis)	Arguments for regulation / intervention e.g. - Market failure - Negative externalities - Inability of markets to make long term predictions - Health and safety issues arising from environmental problems - The need for coordinated international action Arguments for market forces e.g. - Government failure - Efficiency - Competition, enterprise, efficiency - Signalling, incentive, rationing functions of price
Evaluation	- Evaluation of policies in general, e.g. command and control vs. market-focused policies - Evaluation of some specific policies, e.g. prohibitions, regulations, vs. pollution permits, carbon off-setting, taxes, subsidies - Discussion of the quotation (perfectly performing markets vs. wonderfully wise governments) - Consideration of why environmental policies might be good for business and also why it is good for business to respect the environment - Advantages and disadvantages of a mixed economy approach - Market failure versus government failure - Need for privatisation to be accompanied by effective regulation - Discussion of the claim about privatised profits and nationalised costs <i>Examiners should note that, for some of the weaker or average candidates, parts of the above represent quite sophisticated evaluation and so it is important to ensure that more basic evaluation is adequately rewarded where a genuine effort has been made to display that skill.</i>
Also give credit for	- Relevant diagrams - An overall judgement on the issues raised.

It will only be possible for candidates to consider a few of the issues and areas noted. They may also discuss issues not mentioned in the table.

[25]

11.

D

[1]

12.

Relevant issues include:

- the nature of the UK energy market, or any other relevant market chosen by the candidate
- perfect competition
- types of imperfect competition
- concept of efficiency in economics
- what is meant by allocative efficiency
- what is meant by productive efficiency
- why profit maximising firms, operating in imperfectly competitive markets, are unlikely to achieve productive and allocative efficiency

The use of relevant diagrams to support the analysis should be taken into account when assessing the quality of the candidate's response to the question.

[15]

13.

In this part of the question, candidates will need to demonstrate that they are able to evaluate issues and arguments to support a conclusion if they are to be awarded **more than 15 marks**.

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Level 1	Very weak	0 to 3 <i>Mid-Point 2 marks</i>

Issues and areas for discussion include

Introduction	Basic definitions / descriptions of • long run • profit maximisation • business goals / objectives
Developing the response to the question: Application/Analysis	• The importance of the profit motive. • Relevance of the size of firms • The functions of the entrepreneur: ownership, control, decision making and risk taking in small firms and large firms • Directors versus managers • Decision-making in small firms and large firms • The impact of a divorce between ownership and control • Other possible objectives ◦ Sales maximisation ◦ Revenue / turnover maximisation ◦ Maximisation of Market share ◦ Maximisation of company size (world domination) ◦ Industry-specific objectives e.g. newspaper circulation, TV viewing figures ◦ Managerial utility ◦ Social welfare maximisation ◦ Satisficing
Evaluation	• Critical discussion of the quotation • Bonus culture • Short run vs long run • Large firms vs small firms • Overall view of the various possible objectives • Is any one objective sufficient, or do firms need a range of objectives? • Importance of profits for investment • Can any other objectives be achieved in the long run if profits are not achieved? • Is profit maximising actually possible? Do firms have the information? Can they track MC, MR etc? <i>Examiners should note that, for some of the weaker or average candidates, parts of the above represent quite sophisticated evaluation and so it is important to ensure that more basic evaluation is adequately rewarded where a genuine effort has been made to display that skill.</i>
Also give credit for	• Diagrams • An overall judgement on the issues raised • Examples

It will only be possible for candidates to consider a few of the above. They may also discuss issues not discussed above.

Level of response	An answer that:	Max 9 marks
3	- is well organised and develops one or more of the key issues that are relevant to the question - shows sound knowledge and understanding of relevant economic terminology, concepts and principles - includes good application of relevant economic principles and/or good use of data to support the response - includes well-focused analysis with a clear, logical chain of reasoning - includes a relevant diagram that will, at the top of this level, be accurate and used appropriately.	7-9 marks
2	- includes one or more issues that are relevant to the question - shows reasonable knowledge and understanding of economic terminology, concepts and principles but some weaknesses may be present - includes reasonable application of relevant economic principles and/or data to the question - includes some reasonable analysis but it might not be adequately developed and may be confused in places - may include a relevant diagram.	4-6 marks
1	- is very brief and/or lacks coherence - shows some limited knowledge and understanding of economic terminology, concepts and principles but some errors are likely - demonstrates very limited ability to apply relevant economic principles and/or data to the question - may include some very limited analysis but the analysis lacks focus and/or becomes confused - may include a relevant diagram but the diagram is not used and/or is inaccurate in some respects.	1-3 marks

A monopoly diagram is expected, illustrating that where there is collusion, output is likely to be lower and prices higher than under competitive market conditions.

Relevant issues include:

- the nature of collusion
- the ability to control the market and exercise monopoly power
- restricting output to force up the price
- maximising joint profits
- producing where $MC=MR$ to maximise profits
- why competition between suppliers helps to drive prices down

15.

Areas for discussion include:

- roads as a scarce resource
- the negative externalities associated with congestion, eg the longer journey times for other road users
- what is meant by road-pricing and how it might be implemented (allow both tolls and congestion charges as forms of road-pricing or as alternative policies)
- other policies that might be employed such as: higher taxes on fuels, increasing the road fund licence, building more roads, subsidising public transport, investing in and improving public transport
- the extent to which roads display the characteristics of a public good and how has this been affected by developments in technology.
- congestion is most prevalent on certain roads and at certain times of the day
- how road-pricing might be implemented, eg charging drivers a price for each mile travelled with different prices for different roads and for different times of the day, perhaps mentioning automatic number plate recognition systems
- setting up tolls on major roads with the toll charge varying between peak and off-peak times
- how higher taxes on fuels would help to reduce congestion and the relevance of price elasticity of demand
- how subsidising, for example, bus and rail transport, will help to reduce congestion and the relevance of cross elasticity of demand
- the budgetary and cost implications of the different options
- possible unintended consequences of the different methods, eg the impact of road pricing on city centres, such as empty shops, higher prices for products as transport costs rise
- equity arguments, eg are indirect taxes regressive, should the road user pay or the taxpayer, the impact of the different policies on the poorest members of society, town versus country dwellers, commuters and businesses
- the likelihood of government failure, eg inadequate information means that the government may set an inappropriate price for using roads, eg a price that doesn't reflect the true cost of congestion, government policy is likely to be affected by public opinion and the reactions of voters.

The use of relevant diagrams to support the analysis should be taken into account when assessing the quality of the candidate's response to the question.

[25]

16.

Areas for discussion include:

- the nature and examples of price discrimination
- allows the producer to convert consumer surplus into producer surplus
- how price discrimination allows the producer to increase revenue and profit compared to a single price
- relevance of price elasticity of demand
- how price discrimination enables the producer to reduce fixed costs by spreading demand more evenly, eg peak and off-peak rail fares
- better capacity utilisation by firms will lower costs and may lead to lower prices for consumers
- why price discrimination may increase total sales and generate economies of scale
- the use of price discrimination as a barrier to entry, reducing competition
- why price discrimination almost certainly means that some consumers will face a higher price than if there was a single price
- the use of price discrimination to cross-subsidise loss-making services
- some consumers will benefit from lower prices and products that might not otherwise be supplied
- equity considerations related to the redistribution of income that results from price discrimination.

The use of relevant diagrams to support the analysis should be taken into account when assessing the quality of the candidate's response to the question.

The use of examples to illustrate price discrimination should also be taken into account when assessing the quality of response.

[25]

17.

Relevant issues include:

- reasons why prices may differ, eg cost differences, price discrimination
- examples of price discrimination
- the ability of the producer to exert control over supply
- the ability to separate different markets, eg through time, barriers to entry
- preventing the transfer of goods or services between markets
- the significance of different price elasticities of demand
- why cost differences, eg transport costs, may lead to price differences
- why imperfect information may allow firms to charge different prices for the same product
- why product differentiation and branding may allow a supplier to charge different prices for what is essentially the same product.

The use of relevant diagrams to support the analysis should be taken into account when assessing the quality of the candidate's response to the question.

The use of examples to illustrate price discrimination should also be taken into account when assessing the quality of response.

[15]

18.

Response:	Max 4 marks
- includes evidence that shows that energy companies have exploited their market power - clearly explains how this data is evidence of energy companies exploiting their market power	4 marks
- includes evidence that shows that energy companies have exploited their market power - unclear explanation of how this data is evidence of energy companies exploiting their market power	3 marks
- includes evidence that shows that energy companies have exploited their market power - limited explanation of how this data is evidence of energy companies exploiting their market power	2 marks
- includes evidence that does not clearly show that energy companies have exploited their market power - no explanation of how this data is evidence of energy companies exploiting their market power.	1 mark

Relevant issues include:

- Evidence of profits rising by a higher proportion than the rise in wholesale costs is an indication of the exploitation of market power
- Evidence of profits rising by a higher proportion than household bills is an indication of the exploitation of market power.
- the data shows profit element in household bills rose from £10 to £95 between 2009 and 2013
- average bills per household rose from £1145 to £1420 over the period, a larger increase than the rise in wholesale energy element in household bills, £615 to £635
- between 2009 and 2010, wholesale energy costs fell from £615 to £485 per bill but average bills had a smaller decrease, £1145 to £1105
- between 2012 and 2013, wholesale costs increased by £10 per bill at a time when household bills rose by £120 and profits by £45 per bill

[4]

**21.**

For relevant definitions , e.g. price discrimination, in terms of different prices charged to different customers (2 marks) for the same product (2 marks) without the difference explained by cost differences (2 marks) .	Up to 2 marks per point defined or explained to a maximum of 4 marks
For explanation – award 1 mark for each link in a logical chain of reasoning to a maximum of 15 marks.	
Possible explanations include - Some element of monopoly power (1 mark). The supplier needs some ability to make rather than take prices. (1 mark). It must be possible to identify different markets (1 mark) and keep markets separate (1 mark) to prevent seepage (re-selling) between markets (1 mark). Each market has a different price elasticity of demand (1 mark), this allows the supplier to charge whatever price their market will bear (1 mark). - Methods of separating markets, e.g. - Distance - Time - Type of customer - First, second, third degree price discrimination. (Note: these can be rewarded, if discussed, but they are not essential, as they are not required by the specification).	
Use of diagrams to support the explanation, e.g. supply curves with associated demand (AR, MR) curves of differing elasticity	Up to 2 marks per diagram (1 mark for labelling, 1 mark for information shown) to a total of 4 marks.
Relevant real world examples and/or relevant reference to the UK and/or other economies.	1 mark per reference to a maximum of 2 marks

[15]**22.**

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Level 1	Very weak	0 to 3 <i>Mid-Point 2 marks</i>

Issues and areas for discussion include

Introduction	- Definitions of, e.g. - concentration - monopoly power
Developing the response to the question: Application/Analysis	- Advantages of concentration / monopoly - Economies of scale - International competitiveness - Patents, copyrights, etc. - Public interest arguments, e.g. strategic industries, employment issues - Dynamic efficiency - Natural monopoly - Disadvantages - Consumer surplus transferred to producers - Deadweight losses - Higher prices, lower output, under-used capacity - Possibility of collusion - Market failure - Producer sovereignty
Evaluation	- Critical discussion of the quotation - Advantages versus disadvantages of monopoly / concentration - Possibility of price discrimination (positives and negatives) - Need for government intervention and regulation - Collusion distinguished from cooperation / collaboration - Static / dynamic efficiency - Is perfect competition always 'perfect'? - Does 'perfect' necessarily signal 'the best'? - Global issues (large firms more resilient than small ones faced with international competition and corporatism) - Advantages of monopoly in certain situations (e.g. NHS vs health markets) - Application to particular industries, such as motor manufacturing <i>Examiners should note that, for some of the weaker or average candidates, parts of the above represent quite sophisticated evaluation and so it is important to ensure that more basic evaluation is adequately rewarded where a genuine effort has been made to display that skill.</i>
Also give credit for	- Diagrams - An overall judgement on the issues raised - Examples

It will only be possible for candidates to consider a few of the issues and areas noted. They may also discuss issues not mentioned in the table.

[25]

23.

In this part of the question, candidates will need to demonstrate that they are able to evaluate issues and arguments to support a conclusion if they are to be awarded **more than 15 marks**.

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Issues and areas for discussion include

Introduction	- Definitions, examples - Economies of scale - What is meant by “small” or “large” in this context
Developing the response to the question: Application/Analysis	- Distinction between short run and long run - Diminishing returns and returns to scale - Advantages of large firms, e.g. - Types of economy of scale - Other advantages of size, e.g. opportunities for collusion, price making, monopoly, oligopoly, dominant firm price leadership - Disadvantages: diseconomies of scale; x inefficiency - Globalisation and concentrated markets - Government policy (eg sweetheart tax deals) - Advantages of small firms, e.g. - Niche markets - Barometric price leadership - Disruption, e.g. easyJet, Uber - Government policy favouring small firms (eg small business support, start-up grants; business advice agencies) - Government policy hindering small firms, e.g. business rates, VAT, red tape
Evaluation	- Ways in which small firms can fight back, e.g - Reducing costs (internet / weightless economy) - Technical change reducing the need for scale economies (eg nuclear power stations versus windmills) - The importance of small firms in the economy - As employers - As motors for growth - As innovators - As ethical producers - Serving specialised markets - Having the potential for growth – large firms start out small - Arguments for and against government intervention to support / promote Small & Medium Enterprises - The question of whether government intervention is necessary - Overall view of advantages and disadvantages of large and small firms <i>Examiners should note that, for some of the weaker or average candidates, parts of the above represent quite sophisticated evaluation and so it is important to ensure that more basic evaluation is adequately rewarded where a genuine effort has been made to display that skill.</i>
Also give credit for	- Diagrams - An overall judgement on the issues raised - Examples

It will only be possible for students to consider a few of the above. They may also discuss issues not discussed above.

[25]

24.

For relevant definitions, e.g. perfect competition (2 marks), efficient resource allocation (2 marks)	Up to 2 marks per point defined or explained to a maximum of 4 marks
For explanation – award 1 mark for each link in a logical chain of reasoning to a maximum of 15 marks.	
Examples include - Economic theory predicts that under the conditions perfect competition the market will move towards a long run equilibrium (1 mark) where firms are earning only normal profits (1 mark) and there is no incentive (1 mark) for existing firms to leave (1 mark) or new firms to enter (1 mark) the industry. At that point $P = MC = MR = AC = AR$ (1 mark). If price equals marginal cost, the producer is compensated for putting the resources together to produce a good or service (1 mark). If price also represents the marginal benefit, then consumers believe a purchase compensates them for the hard-earned money they spend (1 mark). If marginal cost = marginal benefit neither consumers nor producers are benefitting at the expense of the other (1 mark) while consumers are receiving the goods and services they want (1 mark). This gives allocative efficiency (1 mark). If $MC = AC$, then AC must be at its lowest point and this gives productive efficiency (1 mark). So profit maximisation ($MC = MR$) (1 mark) by a large number of small firms has led to an efficient allocation of resource (1 mark). - Because they do not act as price-takers, and AR is therefore not equal to MR firms in any imperfect form of competition cannot achieve all five elements of $P = MC = MR = AC = AR$ at a single equilibrium point and in that sense is less efficient than perfect competition For students who do not go directly to the LR equilibrium position, but start with the short run, distinguish between firm and industry, and proceed through the 'steps' towards achieving LR equilibrium: - Allow up to 2 marks for a mere 'list' of conditions of perfect competition. Individual conditions can be credited (1 mark each) only if they are discussed within a logical chain of reasoning that explains price-taking and competitive behaviour.	
Use of diagrams to support explanations e.g., Perfect competition, equilibrium, short / long run	Up to 2 marks per diagram (1 mark for labelling, 1 mark for information shown) to a total of 4 marks.
Relevant real world examples and / or relevant reference to the UK and / or other economies.	1 mark per reference to a maximum of 2 marks

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Issues and areas for discussion include

Introduction	Basic definitions/descriptions, e.g. producer surplus/ consumer surplus; regulation
Developing the response to the question: Application / Analysis	- Advantages of price discrimination, e.g. - Producers receive higher profits which, if re-invested, might benefit consumers through higher efficiency - Increased affordability for consumers paying the lower prices (the low cost airline model, which has been replicated by some other industries) - PD might turn a loss making firm into a profitable one - PD might enable a good / service to be produced that otherwise would be lost - Disadvantages - PD turns consumer surplus into producer surplus, so represents a transfer of resources from a less powerful group to a more powerful one (re-distribution of income in reverse) - In extreme cases, everyone pays the maximum they are willing to pay; consumer surplus disappears - Consumers paying the highest prices might switch to goods / services that are less socially optimal, e.g. consumers paying peak-time public transport fares who can afford to switch to private transport might be incentivised to do so.
Evaluation	- Critical discussion of the quotation - Overall consideration of advantages vs. disadvantages - The case for and against regulation - The spread of price discrimination from low-cost airlines and railway journeys to other business models, e.g. hotels, taxi journeys, and according to the data, theme parks and cruise ships: is this good or bad for consumers / producers? - Discussion of the word 'always' in the question <i>Examiners should note that, for some of the weaker or average candidates, parts of the above represent quite sophisticated evaluation and so it is important to ensure that more basic evaluation is adequately rewarded where a genuine effort has been made to display that skill.</i>
Also give credit for	- Diagrams - An overall judgement on the issues raised - Examples

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